



# SAFETY DATA SHEET

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Revision Date 19-Aug-2020  
Version 2

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### Product Identifier

Nama Produk **Crystal Violet (250ml bottle)**  
Product Description: **Crystal Violet (250ml bottle)**  
Cat No. : TM3100B250

### Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### Details of the supplier of the safety data sheet

Company Thermo Scientific Microbiology Sdn Bhd  
No.6, Jalan TTC 6, Taman Teknologi Cheng,Cheng,  
75250 Melaka, Malaysia  
+606 334 0975.

Supplier Oxoid Ltd.  
Wade Road  
Basingstoke, Hants, UK  
RG24 8PW  
Telephone: +44 (0) 1256  
841144.

E-mail address mbd-sds@thermofisher.com

Emergency Telephone Number  
+606 334 0975

## SECTION 2: HAZARDS IDENTIFICATION

### Classification of the substance or mixture

|                                   |                   |
|-----------------------------------|-------------------|
| Flammable liquids                 | Category 3 (H226) |
| Serious Eye Damage/Eye Irritation | Category 2 (H319) |
| Carcinogenicity                   | Category 2 (H351) |
| Chronic aquatic toxicity          | Category 3 (H412) |

### Label Elements



Signal Word

Warning

### Hazard Statements

H226 - Flammable liquid and vapor  
H319 - Causes serious eye irritation

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H351 - Suspected of causing cancer  
H412 - Harmful to aquatic life with long lasting effects

## Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P308 + P313 - IF exposed or concerned: Get medical advice/attention

## Other Hazards

No information available

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component           | CAS-No   | Weight % |
|---------------------|----------|----------|
| C.I. Basic violet 1 | 548-62-9 | 1        |
| Ethyl alcohol       | 64-17-5  | 10       |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. . Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Water mist may be used to cool closed containers.

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## Extinguishing media which must not be used for safety reasons

No information available.

## Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

## Hazardous Combustion Products

None under normal use conditions.

## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

Do not flush into surface water or sanitary sewer system.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component     | Malaysia | ACGIH TLV      | OSHA PEL                                                         |
|---------------|----------|----------------|------------------------------------------------------------------|
| Ethyl alcohol |          | STEL: 1000 ppm | (Vacated) TWA: 1000 ppm<br>(Vacated) TWA: 1900 mg/m <sup>3</sup> |

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|  |  |  |                                              |
|--|--|--|----------------------------------------------|
|  |  |  | TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup> |
|--|--|--|----------------------------------------------|

| Component     | European Union | The United Kingdom                                                                                            | Germany                                           |
|---------------|----------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Ethyl alcohol |                | TWA: 1000 ppm TWA; 1920 mg/m <sup>3</sup><br>TWA<br>WEL - STEL: 3000 ppm STEL;<br>5760 mg/m <sup>3</sup> STEL | 200 ppm TWA MAK; 380 mg/m <sup>3</sup><br>TWA MAK |

## Exposure Controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g.

sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### Appearance

#### Physical State

Liquid

#### Odor

No information available

#### Odor Threshold

No data available

#### pH

No information available

#### Melting Point/Range

No data available

#### Softening Point

No data available

#### Boiling Point/Range

No information available

#### Flash Point

49 °C / 120.2 °F

Method - No information available

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|                                         |                          |                                        |
|-----------------------------------------|--------------------------|----------------------------------------|
| Evaporation Rate                        | No data available        |                                        |
| Flammability (solid,gas)                | Not applicable           | Liquid                                 |
| Explosion Limits                        | No data available        |                                        |
| Vapor Pressure                          | No data available        |                                        |
| Vapor Density                           | No data available        | (Air = 1.0)                            |
| Specific Gravity / Density              | No data available        |                                        |
| Bulk Density                            | Not applicable           | Liquid                                 |
| Water Solubility                        | No information available |                                        |
| Solubility in other solvents            | No information available |                                        |
| Partition Coefficient (n-octanol/water) |                          |                                        |
| Component                               | log Pow                  |                                        |
| C.I. Basic violet 1                     | 0.51                     |                                        |
| Ethyl alcohol                           | -0.32                    |                                        |
| Autoignition Temperature                | No data available        |                                        |
| Decomposition Temperature               | No data available        |                                        |
| Viscosity                               | No data available        |                                        |
| Explosive Properties                    | No information available | explosive air/vapour mixtures possible |
| Oxidizing Properties                    | No information available |                                        |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

### Conditions to Avoid

Keep away from open flames, hot surfaces and sources of ignition.

### Incompatible Materials

None known.

### Hazardous Decomposition Products

None under normal use conditions.

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## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Acute Toxicity

#### Toxicology data for the components

| Component           | LD50 Oral                 | LD50 Dermal | LC50 Inhalation       |
|---------------------|---------------------------|-------------|-----------------------|
| C.I. Basic violet 1 | LD50 = 420 mg/kg ( Rat )  |             |                       |
| Ethyl alcohol       | LD50 = 7060 mg/kg ( Rat ) |             | 20000 ppm/10H ( Rat ) |

#### Chronic Toxicity

##### Carcinogenicity

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component     | IARC    | UK |
|---------------|---------|----|
| Ethyl alcohol | Group 1 |    |

##### Sensitization

No information available

##### Mutagenic Effects

No information available

##### Reproductive Effects

No information available

##### Developmental Effects

No information available

##### Target Organs

None known.

#### Symptoms

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

The product contains following substances which are hazardous for the environment.  
Contains a substance which is: Toxic to aquatic organisms.

| Component           | Freshwater Fish                                            | Water Flea                                                       | Freshwater Algae                                                         | Microtox                                                                                                    |
|---------------------|------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| C.I. Basic violet 1 |                                                            | EC50 = 0.24 - 5 mg/l, 48 h (Daphnia magna (Water flea)) OECD 202 | EC50 = 0.025 - 0.8 mg/l, 72 h (Pseudokirchneriella subcapitata) OECD 201 |                                                                                                             |
| Ethyl alcohol       | Fathead minnow (Pimephales promelas) LC50 = 14200 mg/l/96h | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h                    | EC50 (72h) = 275 mg/l (Chlorella vulgaris)                               | Photobacterium phosphoreum: EC50 = 34634 mg/L/30 min<br>Photobacterium phosphoreum: EC50 = 35470 mg/L/5 min |

### Persistence and degradability

No information available

| Component                             | Degradability |
|---------------------------------------|---------------|
| C.I. Basic violet 1<br>548-62-9 ( 1 ) | 10 %          |

#### Degradation in sewage

Contains substances known to be hazardous to the environment or not degradable in waste

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treatment plant water treatment plants.

**Bioaccumulative potential** No information available

| Component           | log Pow | Bioconcentration factor (BCF) |
|---------------------|---------|-------------------------------|
| C.I. Basic violet 1 | 0.51    | No data available             |
| Ethyl alcohol       | -0.32   | No data available             |

**Mobility in soil** No information available.

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

### **Other Information**

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Can be landfilled or incinerated, when in compliance with local regulations Do not let this chemical enter the environment Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

**Road and Rail Transport** Not regulated

**IATA** Exempt under special provision A58: An aqueous solution containing 24% or less alcohol by volume is not subject to these Regulations.

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component           | EINECS    | ELINCS | NLP | TSCA | DSL | NDSL | PICCS | ENCS | IECSC | AICS | KECL         |
|---------------------|-----------|--------|-----|------|-----|------|-------|------|-------|------|--------------|
| C.I. Basic violet 1 | 208-953-6 | -      |     | X    | X   | -    | X     | X    | X     | X    | KE-0700<br>6 |
| Ethyl alcohol       | 200-578-6 | -      |     | X    | X   | -    | X     | X    | X     | X    | KE-1321<br>7 |

| Component | Seveso III Directive | Seveso III Directive | Rotterdam Convention | Basel Convention |
|-----------|----------------------|----------------------|----------------------|------------------|
|-----------|----------------------|----------------------|----------------------|------------------|

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|               | (2012/18/EC) - Qualifying Quantities for Major Accident Notification | (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | (PIC) | (Hazardous Waste) |
|---------------|----------------------------------------------------------------------|---------------------------------------------------------------------|-------|-------------------|
| Ethyl alcohol |                                                                      |                                                                     |       | Annex I - Y42     |

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** (volatile organic compound)

### Key literature references and sources for data

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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Revision Summary

Initial Release.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**